

Environment Diagrams

Q1: Nested Calls Diagrams

Draw the environment diagram that results from executing the code below.

[See the web version of this resource for the environment diagram.](#)

HOFs

Q2: Currying

Write a function `curry` that will curry any two argument function.

```
def curry(func):
    """
    Returns a Curried version of a two-argument function FUNC.
    >>> from operator import add, mul, mod
    >>> curried_add = curry(add)
    >>> add_three = curried_add(3)
    >>> add_three(5)
    8
    >>> curried_mul = curry(mul)
    >>> mul_5 = curried_mul(5)
    >>> mul_5(42)
    210
    >>> curry(mod)(123)(10)
    3
    """
    def first(arg1):
        def second(arg2):
            return func(arg1, arg2)
        return second
    return first

# Solution with lambda
def curry_lambda(func):
    return lambda arg1: lambda arg2: func(arg1, arg2)
```

First, try implementing `curry` with `def` statements. Then attempt to implement `curry` in a single line using lambda expressions.

Recursion

Q3: Subsequences

A subsequence of a sequence S is a subset of elements from S , in the same order they appear in S . Consider the list $[1, 2, 3]$. Here are a few of its subsequences $[], [1, 3], [2]$, and $[1, 2, 3]$.

Write a function that takes in a list and returns all possible subsequences of that list. The subsequences should be returned as a list of lists, where each nested list is a subsequence of the original input.

In order to accomplish this, you might first want to write a function `insert_into_all` that takes an item and a list of lists, adds the item to the beginning of each nested list, and returns the resulting list.

```
def insert_into_all(item, nested_list):
    """Return a new list consisting of all the lists in nested_list,
    but with item added to the front of each. You can assume that
    nested_list is a list of lists.

    >>> nl = [], [1, 2], [3]
    >>> insert_into_all(0, nl)
    [[0], [0, 1, 2], [0, 3]]
    """
    return [[item] + lst for lst in nested_list]

def subseqs(s):
    """Return a nested list (a list of lists) of all subsequences of S.
    The subsequences can appear in any order. You can assume S is a list.

    >>> seqs = subseqs([1, 2, 3])
    >>> sorted(seqs)
    [], [1], [1, 2], [1, 2, 3], [1, 3], [2], [2, 3], [3]
    >>> subseqs([])
    []
    """
    if not s:
        return []
    else:
        subset = subseqs(s[1:])
        return insert_into_all(s[0], subset) + subset
```

OOP

Q4: Bear

Implement the `SleepyBear` and `WinkingBear` classes so that calling their `print` method matches the doctests. Use as little code as possible and try not to repeat any logic from `Eye` or `Bear`. Each blank can be filled with just two short lines.

```

class Eye:
    """An eye.

    >>> Eye().draw()
    '0'
    >>> print(Eye(False).draw(), Eye(True).draw())
    0 -
    """
    def __init__(self, closed=False):
        self.closed = closed

    def draw(self):
        if self.closed:
            return '-'
        else:
            return '0'

class Bear:
    """A bear.

    >>> Bear().print()
    ? 0o0?
    """
    def __init__(self):
        self.nose_and_mouth = 'o'

    def next_eye(self):
        return Eye()

    def print(self):
        left, right = self.next_eye(), self.next_eye()
        print('? ' + left.draw() + self.nose_and_mouth + right.draw() + '?')

```

```

class SleepyBear(Bear):
    """A bear with closed eyes.

    >>> SleepyBear().print()
    ? -o-?
    """
    def next_eye(self):
        return Eye(True)

class WinkingBear(Bear):
    """A bear whose left eye is different from its right eye.

    >>> WinkingBear().print()
    ? -o0?
    """
    def __init__(self):
        super().__init__()
        self.eye_calls = 0

    def next_eye(self):
        self.eye_calls += 1
        return Eye(self.eye_calls % 2)

```

Linked Lists

Q5: Linear Sublists

Definition: A *sublist* of linked list s is a linked list of some of the elements of s in order. For example, $\langle 3\ 6\ 2\ 5\ 1\ 7 \rangle$ has sublists $\langle 3\ 2\ 1 \rangle$ and $\langle 6\ 2\ 7 \rangle$ but not $\langle 5\ 6\ 7 \rangle$. A *linear sublist* of a linked list of numbers s is a sublist in which the difference between adjacent numbers is always the same. For example $\langle 2\ 4\ 6\ 8 \rangle$ is a linear sublist of $\langle 1\ 2\ 3\ 4\ 6\ 9\ 1\ 8\ 5 \rangle$ because the difference between each pair of adjacent elements is 2.

Implement `linear` which takes a linked list of numbers s (either a `Link` instance or `Link.empty`). It returns the longest linear sublist of s . If two linear sublists are tied for the longest, return either one.

```

def linear(s):
    """Return the longest linear sublist of a linked list s.

    >>> s = Link(9, Link(4, Link(6, Link(7, Link(8, Link(10))))))
    >>> linear(s)
    Link(4, Link(6, Link(8, Link(10))))
    >>> linear(Link(4, Link(5, s)))
    Link(4, Link(5, Link(6, Link(7, Link(8))))
    >>> linear(Link(4, Link(5, Link(4, Link(7, Link(3, Link(2, Link(8)))))))
    Link(5, Link(4, Link(3, Link(2))))
    """
    def complete(first, rest):
        "The longest linear sublist of Link(first, rest) with difference d that starts
        with first."
        if rest is Link.empty:
            return Link(first, rest)
        elif rest.first - first == d:
            return Link(first, complete(rest.first, rest.rest))
        else:
            return complete(first, rest.rest)
    if s is Link.empty:
        return s
    longest = Link(s.first) # The longest linear sublist found so far
    while s is not Link.empty:
        t = s.rest
        while t is not Link.empty:
            d = t.first - s.first
            candidate = Link(s.first, complete(t.first, t.rest))
            if length(candidate) > length(longest):
                longest = candidate
            t = t.rest
        s = s.rest
    return longest

def length(s):
    if s is Link.empty:
        return 0
    else:
        return 1 + length(s.rest)

```

Trees

Q6: Long Paths

Implement `long_paths`, which returns a list of all *paths* in a tree with length at least `n`. A path in a tree is a list of node labels that starts with the root and ends at a leaf. Each subsequent element must be from a label of a branch of the previous value's node. The *length* of a path is the number of edges in the path (i.e. one less than the number

of nodes in the path). Paths are ordered in the output list from left to right in the tree. See the doctests for some examples.

```

def long_paths(t, n):
    """Return a list of all paths in t with length at least n.

    >>> long_paths(Tree(1), 0)
    [[1]]
    >>> long_paths(Tree(1), 1)
    []
    >>> t = Tree(3, [Tree(4), Tree(4), Tree(5)])
    >>> left = Tree(1, [Tree(2), t])
    >>> mid = Tree(6, [Tree(7, [Tree(8)]), Tree(9)])
    >>> right = Tree(11, [Tree(12, [Tree(13, [Tree(14)]))])])
    >>> whole = Tree(0, [left, Tree(13), mid, right])
    >>> print(whole)
    0
      1
        2
          3
            4
            4
            5
      13
      6
      7
      8
      9
    11
    12
      13
        14
    >>> for path in long_paths(whole, 2):
    ...     print(path)
    ...
    [0, 1, 2]
    [0, 1, 3, 4]
    [0, 1, 3, 4]
    [0, 1, 3, 5]
    [0, 6, 7, 8]
    [0, 6, 9]
    [0, 11, 12, 13, 14]
    >>> for path in long_paths(whole, 3):
    ...     print(path)
    ...
    [0, 1, 3, 4]
    [0, 1, 3, 4]
    [0, 1, 3, 5]
    [0, 6, 7, 8]
    [0, 11, 12, 13, 14]
    >>> long_paths(whole, 4)
    [[0, 11, 12, 13, 14]]

```

Note: This worksheet is a problem bank—most TAs will not cover all the problems in discussion section.

```

if n <= 0 and t.is_leaf():
    return [[t.label]]

```

Iterators & Generators

Q7: Something Different

Implement `differences`, a generator function that takes `t`, a non-empty iterator over numbers. It yields the differences between each pair of adjacent values from `t`. If `t` iterates over a positive finite number of values `n`, then `differences` should yield `n-1` times.

```
def differences(t):
    """Yield the differences between adjacent values from iterator t.

    >>> list(differences(iter([5, 2, -100, 103])))
    [-3, -102, 203]
    >>> next(differences(iter([39, 100])))
    61
    """
    last_x = next(t)
    for x in t:
        yield x - last_x
        last_x = x
```

SQL

Q8: A Secret Message

A substitution cipher replaces each word with another word in a table in order to encrypt a message. To decode an encrypted message, replace each word `x` with its corresponding `y` in a code table.

Write a select statement to decode the **original** message *It's The End* using the **code** table.

```
CREATE TABLE original AS
  SELECT 1 AS n, "It's" AS word UNION
  SELECT 2      , "The"      UNION
  SELECT 3      , "End";

CREATE TABLE code AS
  SELECT "Up" AS x, "Down" AS y UNION
  SELECT "Now"      , "Home" UNION
  SELECT "It's"     , "What" UNION
  SELECT "See"      , "Do" UNION
  SELECT "Can"      , "See" UNION
  SELECT "End"      , "Now" UNION
  SELECT "What"     , "You" UNION
  SELECT "The"      , "Happens" UNION
  SELECT "Love"     , "Scheme" UNION
  SELECT "Not"      , "Mess" UNION
  SELECT "Happens", "Go";

SELECT y FROM original, code WHERE word=x ORDER BY n;
```


What happens now? Write another select statement to decode this encrypted message using the same `code` table.

```
CREATE TABLE original AS
  SELECT 1 AS n, "It's" AS word UNION
  SELECT 2      , "The"      UNION
  SELECT 3      , "End";

CREATE TABLE code AS
  SELECT "Up" AS x, "Down" AS y UNION
  SELECT "Now"      , "Home" UNION
  SELECT "It's"     , "What" UNION
  SELECT "See"      , "Do" UNION
  SELECT "Can"      , "See" UNION
  SELECT "End"      , "Now" UNION
  SELECT "What"     , "You" UNION
  SELECT "The"      , "Happens" UNION
  SELECT "Love"     , "Scheme" UNION
  SELECT "Not"      , "Mess" UNION
  SELECT "Happens", "Go";

SELECT b.y
  FROM original, code AS a, code AS b
 WHERE word=a.x AND a.y=b.x
 ORDER BY n;
```